Q & A From Clinical Reporting with {gtsummary} Workshop

Workshop Date: 2022-08-23

Answered Questions

1. **Q:** How do I save PDFs of slides?

   The website will stay up and you can access the slides; if you want to print it go to the slides in full screen view, convert to pdf mode and print

2. **Q:** How can I see tables in viewer pane?

   Global Options > RMarkdown > show output in Viewer Pane

3. **Q:** Link for the code for Dan's quarto slides?

   https://github.com/ddsjoberg/clinical-reporting-gtsummary-rmed/tree/main/slides

4. **Q:** Syntax for `set_variable_labels()`?

   See function docs at: https://larmarange.github.io/labelled/reference/var_label.html

5. **Q:** How to confirm labels worked?

   `str(df)` or `view(df)`

6. **Q:** What is `|>`?

   It's the new base R pipe! See https://ivelasq.rbind.io/blog/understanding-the-r-pipe/ for more details

7. **Q:** What statistics can you use in `tbl_summary(statistic=)` argument? How does the function know `"{min}, {max}"` is the range?

Characteristic	**N = 200**[1]
Age	47
Unknown	11

[1]Mean

Characteristic	**N (Non-missing)**	**Overall** N = 200[1]	**Drug A** N = 98[1]	**Drug B** N = 102[1]
Age	189	47 (38, 57)	46 (37, 60)	48 (39, 56)
Grade	200			
I		68 (34%)	35 (36%)	33 (32%)
II		68 (34%)	32 (33%)	36 (35%)
III		64 (32%)	31 (32%)	33 (32%)

[1]Median (Q1, Q3); n (%)

There are a set of known names for default statistics that you can use in this function. These (plus glue syntax) allow you to display specific statistics. See more details here, under the `statistic argument` section: https://www.danieldsjoberg.com/gtsummary/reference/tbl_summary.html

4. **Q:** How do you display mean instead of median?

   Adjust the statistic argument:

```
trial %>%
  select(age) %>%
  tbl_summary(statistic = everything() ~ "{mean}")
```

5. **Q:** With `add_n()` can you change the label to specify that it represents non-missing values?

   Yes, you can change this with `col_label` argument:

```
trial |>
  select(age, grade, trt) %>%
  tbl_summary(
    by = trt,
    missing = "no"
  ) |>
  add_overall() |>
  add_n(col_label = "**N (Non-missing)**")
```

Characteristic	**Overall** N = 200[1]	**Drug A** N = 98[1]	**Drug B** N = 102[1]
Age	47 (38, 57)	46 (37, 60)	48 (39, 56)
Grade			
I	68 (34%)	35 (36%)	33 (32%)
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[1]Median (Q1, Q3); n (%)

Characteristic	**Overall** N = 200[1]	**Drug A** N = 98[1]	**Drug B** N = 102[1]
Age	47 (38, 57)	46 (37, 60)	48 (39, 56)
Grade			
I	68 (34%)	35 (36%)	33 (32%)
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[1]Median (Q1, Q3); n (%)

6. **Q:** How can you remove the death variable inside the `tbl_summary()` function?

   Use a`dplyr::select()` prior to tbl_summary or use the `include = argument` in `tbl_summary()`. Three options below:

```
sm_trial <- trial |>
  select(age, grade, trt, death)

sm_trial |>
  select(-death) %>%
  tbl_summary(
    by = trt,
    missing = "no"
  ) |>
  add_overall()
```

```
sm_trial |>
  tbl_summary(
    by = trt,
    include = -death,
    missing = "no"
  ) |>
  add_overall()
```

Characteristic	**Overall** N = 200[1]	**Drug A** N = 98[1]	**Drug B** N = 102[1]
Age	47 (38, 57)	46 (37, 60)	48 (39, 56)
Grade			
I	68 (34%)	35 (36%)	33 (32%)
II	68 (34%)	32 (33%)	36 (35%)
III	64 (32%)	31 (32%)	33 (32%)

[1]Median (Q1, Q3); n (%)

```
sm_trial |>
  tbl_summary(
    by = trt,
    include = !death,
    missing = "no"
  ) |>
  add_overall()
```

7. **Q:** Is it possible to add custom footnotes, say if you want to explain what a particular category includes etc.?

   Yes, use `modify_footnote()` for column headers for footnotes that already exist. Or use `modify_table_styling()` for new footnotes. https://www.danieldsjoberg.com/gtsummary/reference/modify_table_styling.html

8. **Q:** Does gtsummary support role selection like {recipes}?

   Use the selectors `all_continuous()`, `all_categorical()`, etc.

9. **Q:** There is an error trying to use `add_difference()` with `add_p()`

   This is because you are trying to add two p's to same table. You could consider doing two separate tables then using `tbl_merge()` to merge them, but presenting these two p-values in one table could be confusing.

10. **Q:** It would be great to have an "asterisk_p" to show significant p-values, like `bold_p()`

    `add_significance_stars()` already exists! https://www.danieldsjoberg.com/gtsummary/reference/add_significance_stars.html

11. **Q:** If there are more than two groups, is there any function for post-hoc testing?

    Here are supported tests: https://www.danieldsjoberg.com/gtsummary/reference/tests.html. Post-hoc tests might require custom testing. There are some relevant questions on stackoverflow.com that may be useful.

Characteristic	**Beta**	**95% CI**	**p-value**
ECOG Score	-119	-233, -5.5	0.040
Sex	11	-79, 102	0.8
ECOG Score * Sex	43	-33, 120	0.3

Abbreviation: CI = Confidence Interval

12. **Q:** Can gtsummary give Bayesian statistics result?

    See https://www.danieldsjoberg.com/gtsummary/reference/tests.html for supported tests. If it is not supported, you can use custom functions for p-values (see `add_p()`) or `add_stat()` to use a custom statistic.

13. **Q:** The {rms} package fits are not supported?

    You can file an issue for support. You can also use custom tidier. See Wald example in table gallery for tips on how to do this: https://www.danieldsjoberg.com/gtsummary/articles/gallery.html#wald-ci

14. **Q:** How does it handle interaction terms?

    Gorgeously:

```
tbl <-
      lm(time ~ ph.ecog*sex, survival::lung) %>%
      tbl_regression(label = list(ph.ecog = "ECOG Score", sex = "Sex"))

tbl
```

15. **Q:** What does `exponentiate=TRUE` do again in `tbl_regression()`? I missed it.

    Exponentiation coefficients so you can easily report OR instead of log(OR) or HRs instead of raw coefficients.

16. **Q:** Is there a way to add an additional footnote saying T1 is the control? The default dash may be interpreted as a “missing value”.

    `modify_table_styling()` https://www.danieldsjoberg.com/gtsummary/reference/modify_table_styling.html or use argument `add_estimate_to_reference_rows = TRUE`

Characteristic	**OR**	**95% CI**	**p-value**
Age	1.01	0.99, 1.03	0.3
Grade			0.053
I	Ref.	—	
II	1.11	0.56, 2.24	
III	2.28	1.11, 4.75	

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
glm(death ~ age + grade, data = trial, family = binomial) %>%
  tbl_regression( exponentiate = TRUE, add_estimate_to_reference_rows = FALSE ) %>%
  add_global_p() %>%
  modify_table_styling(columns = c(estimate, ci),
                       rows = reference_row == TRUE,
                       missing_symbol = "Ref." )
```

```
Warning: Use of the "ci" column was deprecated in gtsummary v2.0, and the column will
eventually be removed from the tables.
! Review `?deprecated_ci_column()` for details on how to update your code.
i The "ci" column has been replaced by the merged "conf.low" and "conf.high"
  columns (merged with `modify_column_merge()`).
i In most cases, a simple update from `ci = 'a new label'` to `conf.low = 'a
  new label'` is sufficient.
```

You can also use themes and change `tbl_regression-str:ref_row_text` option: https://www.danieldsjoberg.com/gtsummary/articles/themes.html

17. **Q:** Can `add_global_p()` and the global p-value without removing the p-value for each level?

    There is an arg in `add_global_p()` called `keep`; that will keep both p-values. It is `FALSE` by default

```
glm(death ~ age + grade, data = trial, family = binomial) %>%
  tbl_regression(exponentiate = TRUE,
                 add_estimate_to_reference_rows = FALSE) %>%
  add_global_p(keep = TRUE)
```

18. **Q:** `tbl_regression()`: Are the p-values per Wald by default?

Characteristic	OR	95% CI	p-value
Age	1.01	0.99, 1.03	0.3
Grade			0.053
I	—	—	
II	1.11	0.56, 2.24	0.8
III	2.28	1.11, 4.75	0.025

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

You need to look at the details for your specific model and what the summary function returns. You can change default via the tidier for your model. My tidier function example available in the gallery: https://www.danieldsjoberg.com/gtsummary/articles/gallery.html#wald-ci

19. **Q:** Is there an option to compare multiple regression models within a single table? (e.g., when performing sensitivity analyses). I'm thinking of a situation where we are using different adjustment sets of covariates. I'm thinking of a situation where we are using different adjustment sets of covariates.

    Use `tbl_merge()`

20. **Q:** Does it work with {tidymodels}?

    There are currently some issues with dummy variables, but we are working on it for future release.

21. **Q:** Can you have super/sub scripts in variable names

    Use utf8 characters and markdown characters. {gt} sometimes has some issues with "common markdown" language so be aware of your output type and print engine, but most likely they will work.

22. **Q:** Is the marginal adjusted mean estimated using the median of the other covariates fitting the model?

    This is calculated outside the package. See `emmeans::emmeans()` for details on calculations.

23. **Q:** Does `tbl_regression()` support ridge/lasso regressions? I usually use {glmnet} for these

    Not currently supported but see https://github.com/ddsjoberg/gtsummary/issues/1280 for more details.

24. **Q:** For sex, how do i get it to show the `Male` level instead of the `Female` reference group?

Characteristic	Beta	95% CI	p-value
Marker Level (ng/mL)	-0.04	-2.6, 2.5	>0.9
Grade			
I	—	—	
II	0.64	-4.7, 6.0	0.8
III	2.4	-2.8, 7.6	0.4
R^2	0.005		
Adjusted R^2	-0.012		

Abbreviation: CI = Confidence Interval

`inline_text()` has a `level=` argument to select this.

25. **Q:** If I change the CI to 90%, I need to update the pattern to 90% CI {ci} right?

    If you are using the default pattern (`{conf.level}`), you don't need to change anything, but if you hard code like 95%, then yes, use `pattern = "odds ratio {estimate}; {conf.level*100}% CI {ci}"`

26. **Q:** Can I add LaTeX symbols in the inline text like chi square results?

    Probably?

27. **Q:** Can `add_glance_source_note()`, give just adjusted R^2 ?

    Yes!

```
mod <- lm(age ~ marker + grade, trial) %>% tbl_regression()
mod %>%
  add_glance_table(include = c(r.squared, adj.r.squared))
```

28. **Q:** How can I increase the font size of a gtsummary object in a xaringan slide?

    You can set a theme for this, or check out print engine options. See https://www.danieldsjoberg.com/gtsummary/articles/rmarkdown.html

29. **Q:** How do I decrease the height of the gtsummary table so it exports properly?

    Depends on print engine and desired output format. See https://www.danieldsjoberg.com/gtsummary/articles/rmarkdown.html

30. **Q:** What would the code be to export as XLS?

    `as_hux_xlsx()`. See: https://www.danieldsjoberg.com/gtsummary/reference/as_hux_table.html